

PHP JWT Library: RSA1_5 (RSAES-PKCS1-v1_5) decryption lacks implicit rejection, exposing a Bleichenbacher/Marvin padding oracle

Report Type: Weekly · Generated: June 18, 2026 22:39 UTC · Coverage: Jun 18 - Jun 18, 2026

1 Total Advisories	0 Critical	0 High	1 Medium
MEDIUM Severity	N/A CVSS / Risk Score	TLP:CLEAR TLP Classification	PRO Customer Tier

Executive Summary

Impact `RSACrypt::decryptWithRSA15()` (used by the `RSA1\_5` key-encryption algorithm) implements RSAES-PKCS1-v1_5 decryption by inspecting the padding after RSADP and throwing `InvalidArgumentException` as soon as the padding is malformed. It does **not** implement the implicit-rejection countermeasure required by RFC 3447 §7.2.2 / RFC 8017 §7.1.2 (return a deterministic pseudo-random value of the expected length on padding failure and let the downstream step fail uniformly). From a JWE caller this yields a Bleichenbacher/Marvin padding oracle: an attacker submitting adaptively crafted `encrypted\_key` values can distinguish (a) padding rejected, (b) padding valid but wrong CEK length, and (c) padding valid and full AEAD executed — even though `JWEDecrypter` returns the same `false` in all cases — because each path performs a measurably different amount of work, amplifiable by enlarging the ciphertext (CWE-208 timing side channel). Enough adaptive queries can recover the wrapped C

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Top Threat Advisories

High-priority advisories ranked by CVSS score, exploitability, and actor attribution.

Advisory Title	Severity	CVSS	EPSS	AI Summary
PHP JWT Library: RSA1_5 (RSAES-PKCS1-v1_5) decryption lacks implicit r	Critical		-	### Impact `RSACrypt::decryptWithRSA15()` (used by the `RSA1_5` key-enc

MITRE ATT&CK® v15 Mapping

Technique density score: 0.0 · Techniques observed: 0 · Tactics covered: 0

MITRE ATT&CK techniques pending analyst enrichment. Deploy detection rules from Section 6 to identify related activity.

Threat Actor Intelligence

Attribution analysis across advisories in this report period.

Threat Actor	Count	Associated CVEs / Indicators
CDB-UNATTR-CVE	1	-

Indicators of Compromise (IOCs)

IOC extraction for this advisory is pending enrichment pipeline completion. PRO subscribers receive real-time IOC push to SIEM/SOAR via the SENTINEL APEX webhook at: **GET /api/v1/intell/iocs?advisory={id}**. Full IOC packages include IPv4, domain, SHA256, and URL indicators with confidence scores, first-seen timestamps, and STIX 2.1 formatted bundles.

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Detection Engineering

Deploy these production-ready detection rules into your SIEM platform to identify exploitation attempts in real time. Compatible with Microsoft Sentinel, Splunk ES, and any Sigma-compatible platform.

Sigma Rule - Webserver / Process Monitoring

```
title: SENTINEL APEX - CDB-ADVISORY Exploitation Attempt
id: cdb-cdb-advisory-det
status: experimental
description: Detects exploitation indicators related to CDB-ADVISORY.
references:
- https://intel.cyberdudebivash.com
author: CyberDudeBivash SENTINEL APEX
date: 2026/06/18
tags:
- attack.initial_access
- attack.t1190
logsource:
category: webserver
detection:
keywords:
- 'CDB-ADVISORY'
condition: keywords
falsepositives:
- Security testing / penetration testing
level: medium
```

Microsoft Sentinel - KQL Query

```
// SENTINEL APEX - PHP JWT Library: RSA1_5 (RSAES-PKCS1-v1_5) decryption lacks implicit rejection, exposing a
Bleichenbacher/Marvin padding oracle
// Advisory | Microsoft Sentinel KQL
let time_window = ago(24h);
SecurityAlert
| where TimeGenerated > time_window
| where Description has "PHP JWT Library: RSA1_5 (RSAES-PKCS1-v1_"
or AlertName has "PHP JWT Library: RSA1_5 (RSAES-PKCS1-v1_"
or ExtendedProperties has "PHP JWT Library: RSA1_5 (RSAES-PKCS1-v1_"
| extend Rule = "APEX-ADVISORY", Severity = "MEDIUM"
| project TimeGenerated, AlertName, AlertSeverity, Entities, Description, Rule, Severity
| order by TimeGenerated desc
```

Incident Response Playbook

Phase 1 - 0-24 Hours (Containment)

1. Activate IR team; assign lead analyst and executive sponsor.
2. Isolate or WAF-shield PHP JWT Library: RSA1_5 (RSAES instances exposed to the internet.
3. Enable verbose access logging on all PHP JWT Library: RSA1_5 (RSAES endpoints immediately.

- 4. Pull last 72h of web server and application logs for forensic baselining.
- 5. Check SENTINEL APEX IOC feeds and SIEM alerts for exploitation hits in your environment.
- 6. Notify relevant internal stakeholders and legal / compliance team.

Phase 2 - 24-72 Hours (Eradication)

- 1. Apply vendor patch for PHP JWT Library: RSA1_5 (RSAES or implement recommended mitigation).
- 2. Deploy Sigma and KQL detection rules from Section 6 across all SIEM platforms.
- 3. Rotate all API keys, service account credentials, and session tokens.
- 4. Conduct forensic timeline reconstruction for potential compromise window.
- 5. Validate patch deployment across 100% of affected instances via asset inventory.

Phase 3 - 7 Days (Recovery & Hardening)

- 1. Conduct post-incident review and update incident response runbooks.
- 2. Threat hunt for lateral movement or persistence artifacts using MITRE ATT&CK mapping.
- 3. Update asset inventory with patched version information and closure evidence.
- 4. Submit findings to internal risk register and CISO dashboard.
- 5. Schedule follow-up vulnerability scan to confirm remediation completeness.

Financial Impact Analysis (FAIR Model)

\$150K - \$900K Breach Cost Range (FAIR)	\$380K IBM Cost of Breach 2025 Median	\$120K/day Business Interruption Cost	\$85K Regulatory Fine Exposure
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Medium severity events require internal investigation and targeted customer notification. Average cost includes 340 hours of engineering time.

Remediation Cost Estimate: \$55K · Includes engineering time, IR retainer activation, forensic investigation, and customer notification costs.

Regulatory & Compliance Implications

Organizations processing this advisory must evaluate obligations under applicable regulatory frameworks. The table below maps this threat to relevant compliance requirements.

Framework	Reference	Obligation
PCI-DSS v4.0	Req. 6.3.3	Non-critical patches applied within 3 months
NIST CSF 2.0	ID.RA / PR.PS	Vulnerability assessment and platform security policies
ISO 27001:2022	A.8.8	Vulnerability management programme with documented remediation timelines

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Actionable Recommendations

Prioritized defensive actions based on this period's intelligence.

1. Monitor PHP JWT Library: RSA1_5 (RSAES-PKCS1-v1_5) decryption lacks implicit rejection, exposing a Bleichenbacher/Marvin padding oracle for escalation.
2. Apply patches in next maintenance window.
3. Apply vendor patch for PHP JWT Library: RSA1_5 (RSAES-PKCS1-v1_5) decryption lacks implicit rejection, exposing a Bleichenbacher/Marvin padding oracle immediately - check NVD for official fix guidance.
4. Enable enhanced logging and alerting on all systems running affected software versions.
5. Deploy the Sigma detection rule from this report into your SIEM platform (Sentinel / Splunk / Elastic).
6. Audit all internet-facing instances of affected products and confirm patch deployment.
7. Conduct a threat hunt using MITRE ATT&CK techniques mapped in this advisory.

Appendix - Report Metadata & Legal

Field	Value
Report ID	intel--af63dceef5893b6c5ccc6983
Report Type	Weekly
Platform	CYBERDUDEBIVASH SENTINEL APEX v166.0 GOD-MODE
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Customer Tier	PRO
Customer Email	-
TLP	TLP: CLEAR
STIX Version	2.1
ATT&CK Version	v15

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